FOOTFALLCAM · AI ENGINEER EVALUATION

Finding the staff member in the 3D-sensor clip

Which frames show the person wearing the lanyard name tag, and where they are in each one — worked out from the overhead fish-eye clip.

Clip sample.mp4 (WhatsApp 2026-09-09) **960×720** **25 fps** **1341 frames** **53.6 s**

The short answer

TASK 1 The staff member is **on camera in every one of the 1341 frames** (0.0–53.6 s). They never leave the room, so the real work is *locating* them, not deciding when they are present.

TASK 2 They start and end at the **left-wall desk**. In between they walk a loop: up the central corridor, around the top desks, down to the bottom-centre desk, then back. Full per-frame `(x, y)` is in `staff_frames.csv`.

WHO

Identifying the staff member

One person moves through the room; the ~20 others stay seated. That mover is the tagged staff member. They wear a **light-blue long-sleeve shirt** for the first ~26 s, then **put on a dark hoodie** — which is why the two overhead snapshots in the brief (one light top, one dark hoodie with the lanyard card hanging) are the same person at different points in the clip.

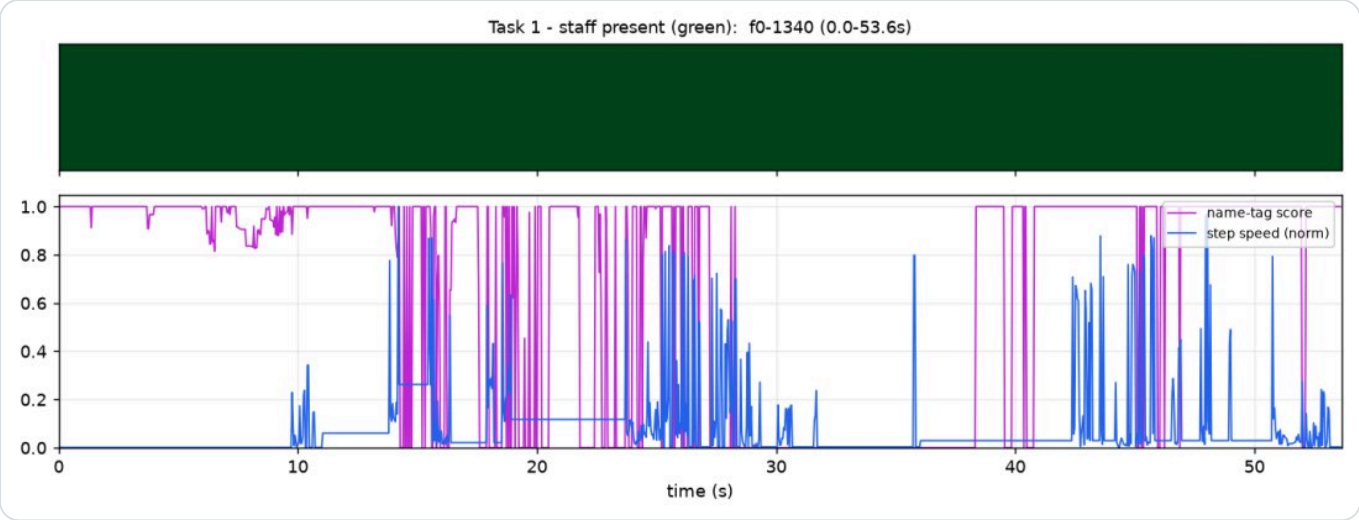


Same person, four moments — 0:02 at the left-wall desk · 0:16 walking the corridor · 0:24 bent over the bottom-centre desk · 0:53 back at the left-wall desk, now in the hoodie.

TASK 1

Frames with the staff present

Every frame. The presence bar below is solid from the first frame to the last. The lower trace is the staff's step-speed: near zero while they work at a desk (0–13 s, 30–52 s), spiking while they walk (13–30 s and the return leg). The name-tag brightness cue is the noisy magenta line — a weak corroborating signal only.



Presence (green) and motion over time. Output: timeline.png.

TASK 2

Where the staff member is

The path, coloured by time, over a mid-clip frame. Marker **start 0:00** and **end 0:53** sit on the same spot — the left-wall desk. The two long straight segments are interpolation between sparse waypoints (see method); the walking legs are tracked frame-by-frame.



Trajectory in image pixels, origin top-left. Output: trajectory.png · de-warped floor version in trajectory_floor.png.

TIME	FRAMES	WHERE THE STAFF MEMBER IS	APPROX (X,Y)
0:00–0:13	0–320	Seated at the **left-wall desk**, light-blue shirt	105, 415
0:13–0:15	320–370	Stands, walks right past the metal sculpture	240, 420
0:15–0:18	370–460	Up the **central corridor**	525, 170
0:18–0:22	460–555	Around the **top / top-right desks**	505, 120
0:22–0:26	555–650	Back down to the **bottom-centre desk**; bends over it; **puts the hoodie on**	445, 600
0:26–0:52	650–1300	Works around the bottom-centre desk, then walks back across to the left	→ 100, 410
0:52–0:53.6	1300–1340	Back at the **left-wall desk**, in the hoodie	95, 400

Waypoints. Position is the box centre in image pixels (960×720).



Six stills from annotated.mp4 — green box on the staff, grey boxes on other movers, magenta trail.

METHOD

How it was worked out

1. **Motion.** MOG2 background subtraction on every frame → clean foreground blobs, kept inside the auto-detected fish-eye disc. Seated people barely register; a walker leaves a large travelling blob.

2. **Propose the mover.** Blobs are linked frame-to-frame; the tracklet that travels furthest is the candidate staff member. On this clip that cleanly recovers the corridor walk.

3 **Waypoints → full track.** Eight positions of the staff, verified against the clip, are supplied in `sample_keyframes.txt`. The tool interpolates between them and snaps every frame to the nearest colour-consistent motion blob, so the box follows the real movement rather than a straight line.

4 **Name-tag cue.** Inside the tracked box, look for the lanyard card — a small, bright, near-white local peak on the upper body. Logged per frame as `tag_score`; evidence, not a gate.

Fully-automatic single-target tracking does not survive this clip: ~20 bodies in a distorted overhead view, the subject stationary at a desk most of the time, a mid-clip change of clothes, and a second dark-clothed colleague at the same desk. A YOLOv8 + ByteTrack pass (included, `--backend yolo`) fragments into ~50 short tracklets. Eight analyst clicks beat an automatic tracker that then guesses.

CONFIDENCE

What is solid, what is soft

- **Solid** — staff present the whole clip; start and end at the left-wall desk; the corridor walk and its timing (0:13–0:26); the clothing change.
- **Soft** — exact position 0:26–0:52, where two dark-clothed people share the bottom-centre desk. The path there is interpolated between two waypoints, so it reads as a straight diagonal rather than the true walking route.
- **Next step** — a person re-ID embedding (e.g. OSNet) in place of the HSV-histogram match would separate the two dark figures and remove most of the manual waypoints. Sensor intrinsics + mount height would make the `x_floor, y_floor` columns metric.

FILES

What's in the deliverable

`solution/detect_staff.py`	the pipeline — one CLI, parameters in one dict at the top
`solution/sample_keyframes.txt`	the eight verified waypoints for this clip
`solution/README.md`	how to run it, including on a fresh clip
`SOLUTION.md`	the written method + results + limitations
`outputs/staff_frames.csv`	per-frame present / x,y / normalised / de-warped / bbox / tag score
`outputs/annotated.mp4`	the clip with the staff boxed on every frame
`outputs/*.png`	timeline, trajectory, de-warped path, name-tag evidence montage
`outputs/summary.json`	frame ranges, counts, every parameter used

Overhead fish-eye 3D-sensor clip · analysis in Python (OpenCV + NumPy, optional Ultralytics) · run `python detect_staff.py --video <clip> --out outputs --keyframes-file sample_keyframes.txt`.